

CSE370 : Database Systems

Assignment 03 | Summer 2026

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No 1 Query (as Plain Text)	select c.customer_name,l.loan_number from customer c inner join borrower b on c.customer_id=b.customer_id inner join loan l on b.loan_number=l.loan_number where l.branch_name = 'Downtown';
No 1 SS (of Query & Output in Shell)	<pre>MariaDB [bank_24301485]> select c.customer_name,l.loan_number from customer c inner join borrower b on c.customer_id=b.c ustomer_id inner join loan l on b.loan_number=l.loan_number where l.branch_name = 'Downtown'; +-----+-----+ customer_name loan_number +-----+-----+ Johnson L-14 Jones L-17 Williams L-17 +-----+-----+ 3 rows in set (0.001 sec)</pre>
No 2 Query (as Plain Text)	select distinct c.customer_name from customer c inner join borrower b on c.customer_id = b.customer_id;
No 2 SS (of Query & Output in Shell)	<pre>MariaDB [bank_24301485]> select distinct c.customer_name from customer c inner join borrower b on c.customer_id = b.cust omer_id; +-----+ customer_name +-----+ Jones Smith Hayes Curry Williams Adams Johnson +-----+ 7 rows in set (0.001 sec)</pre>

No 3 Query (as Plain Text)	select c.customer_name from customer c left join borrower b on c.customer_id=b.customer_id where b.customer_id is null;
No 3 SS (of Query & Output in Shell)	<pre> MariaDB [bank_24301485]> select c.customer_name from customer c left join borrower b on c.customer_id=b.customer_id where b.customer_id is null; +-----+ customer_name +-----+ Lindsay Turner Glenn Brooks Green +-----+ 5 rows in set (0.001 sec) </pre>
No 4 Query (as Plain Text)	select distinct c1.customer_name as Customer1, c2.customer_name as Customer2, c1.customer_city as City from customer c1 inner join customer c2 on c1.customer_city=c2.customer_city where c1.customer_id < c2.customer_id;
No 4 SS (of Query & Output in Shell)	<pre> MariaDB [bank_24301485]> select distinct c1.customer_name as Customer1, c2.customer_name as Customer2, c1.customer_city as City from customer c1 inner join customer c2 on c1.customer_city=c2.customer_city where c1.customer_id < c2.customer_id; +-----+-----+-----+ Customer1 Customer2 City +-----+-----+-----+ Jones Hayes Harrison Smith Curry Rye Lindsay Adams Pittsfield Turner Green Stamford +-----+-----+-----+ 4 rows in set (0.001 sec) </pre>
No 5 Query (as Plain Text)	select branch_name as Branch_Name, round(sum(balance*0.04),2) as Total_Interest from account group by branch_name;

No 5 SS (of Query & Output in Shell)	<pre> MariaDB [bank_24301485]> select branch_name as Branch_Name, round(sum(balance*0.04),2) as Total_Interest from account group by branch_name; +-----+-----+ Branch_Name Total_Interest +-----+-----+ Brighton 66.00 Downtown 20.00 Mianus 28.00 Perryridge 16.00 Redwood 28.00 Round Hill 14.00 +-----+-----+ 6 rows in set (0.001 sec) </pre>
No 6 Query (as Plain Text)	SELECT b.branch_city, a.account_number, a.balance FROM account a JOIN branch b ON a.branch_name = b.branch_name where (b.branch_city, a.balance) in (SELECT b2.branch_city, MAX(a2.balance) from account a2 JOIN branch b2 ON a2.branch_name = b2.branch_name GROUP BY b2.branch_city);
No 6 SS (of Query & Output in Shell)	<pre> MariaDB [bank_24301485]> SELECT b.branch_city, a.account_number, a.balance FROM account a JOIN branch b ON a.branch_name = b.branch_name where (b.branch_cit y, a.balance) in (SELECT b2.branch_city, MAX(a2.balance) from account a2 JOIN branch b2 ON a2.branch_name = b2.branch_name GROUP BY b2.branch_city); +-----+-----+-----+ branch_city account_number balance +-----+-----+-----+ Brooklyn A-201 900 Horseneck A-215 700 Palo Alto A-222 700 +-----+-----+-----+ 3 rows in set (0.001 sec) </pre>
No 7 Query (as Plain Text)	select * from (select l.loan_number, l.amount, c.customer_name from loan l join borrower b on l.loan_number = b.loan_number join customer c on b.customer_id = c.customer_id order by l.amount DESC limit 5) as top_loans order by amount ASC, loan_number DESC;

No 7 SS (of Query & Output in Shell)	<pre> MariaDB [bank_24301485]> select * from (select l.loan_number, l.amount, c.customer_name from loan l join borrower b on l.loan_number = b.loan_number join customer c on b.customer_id = c.customer_id order by l.amount DESC limit 5) as top_loans order by amount ASC, loan_number DESC; +-----+-----+-----+ loan_number amount customer_name +-----+-----+-----+ L-17 1000 Jones L-16 1300 Adams L-15 1500 Hayes L-14 1500 Johnson L-23 2000 Smith +-----+-----+-----+ 5 rows in set (0.001 sec) </pre>
No 8 Query (as Plain Text)	<pre> select distinct c.customer_name from customer c inner join depositor d on c.customer_id = d.customer_id inner join account a on d.account_number = a.account_number inner join borrower b on c.customer_id = b.customer_id inner join loan l on b.loan_number=l.loan_number where a.branch_name='Perryridge' and l.branch_name='Perryridge'; </pre>
No 8 SS (of Query & Output in Shell)	<pre> MariaDB [bank_24301485]> select distinct c.customer_name from customer c inner join depositor d on c.customer_id = d.customer_id inner join account a on d.account_number = a.account_number inner join borrower b on c.customer_id = b.customer_id inner join loan l on b.loan_number=l.loan_number where a.branch_name='Perryridge' and l.branch_name='Perryridge'; +-----+ customer_name +-----+ Hayes +-----+ 1 row in set (0.001 sec) </pre>
No 9 Query (as Plain Text)	<pre> select c.customer_name, SUM(l.amount) as total_loan from customer c join borrower b on c.customer_id = b.customer_id join loan l on b.loan_number = l.loan_number group by c.customer_id, c.customer_name having count(l.loan_number) >= 2; </pre>

No 9 SS (of Query & Output in Shell)	<pre> MariaDB [bank_24301485]> select c.customer_name, SUM(l.amount) as total_loan from customer c join borrower b on c.customer_id = b.customer_id join loan l on b.loan_number = l.loan_number group by c.customer_id, c.customer_name having count(l.loan_number) >= 2; +-----+-----+ customer_name total_loan +-----+-----+ Smith 2900 +-----+-----+ 1 row in set (0.001 sec) </pre>
No 10 Query (as Plain Text)	<pre> select d.customer_id, sum(a.balance) as total_balance from depositor d join account a ON d.account_number = a.account_number group by d.customer_id having count(a.account_number) > 1; </pre>
No 10 SS (of Query & Output in Shell)	<pre> MariaDB [bank_24301485]> select d.customer_id, sum(a.balance) as total_balance from depositor d join account a ON d.account_number = a.account_number group by d.customer_id having count(a.account_number) > 1; +-----+-----+ customer_id total_balance +-----+-----+ C-226 1400 +-----+-----+ 1 row in set (0.001 sec) </pre>